

ADVANCED MATHEMATICS

Stage 6

Statistics (Adv), S2 Interpretation and Bivariate Data (Adv)

Summary Statistics - No graph (Y12)

Other Chart Types (Y12)

Classifying Data (Y12)

Summary Statistics - Box Plots (Y12)

Bivariate Data Analysis (Y12)

Bar Charts and Histograms (Y12)

Statistics (Adv), S3 Random Variables (Adv)

Normal Distribution (Y12)

Probability Density Functions (Y12)

Teacher: Spiro Gouzoulis

Exam Equivalent Time: 123 minutes (based on allocation of 1.5 minutes per mark)

Questions

1. Statistics, 2ADV S2 2022 HSC 2 MC

⚡ RAP Data (vs state mean) - Bottom 13%: School result (86%) was 3% below state average (89%)

Consider the following dataset.

13 16 17 17 21 24

Which row of the table shows how the median and mean are affected when a score of 5 is added to the dataset?

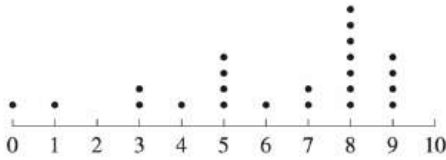
	Median	Mean
A.	Changes	Changes
B.	Stays the same	Stays the same
C.	Changes	Stays the same
D.	Stays the same	Changes

2. Statistics, STD2 S1 2010 HSC 1 MC

⚡ RAP Data (vs state mean) - Bottom 11%: School result (63%) was 13% below state average (76%)

The results of a survey are displayed in the dot plot.

What is the range of this data?



- (A) 7
- (B) 8
- (C) 9
- (D) 10

3. Statistics, STD2 S1 2009 HSC 3 MC

⚡ RAP Data (vs state mean) - Bottom 17%: School result (66%) was 9% below state average (75%)

The eye colours of a sample of children were recorded.

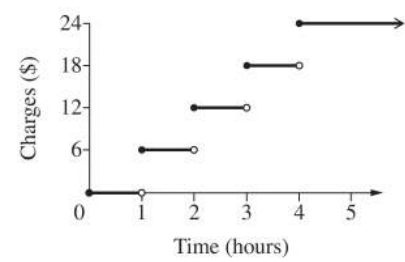
When analysing this data, which of the following could be found?

- A. Mean
- B. Median
- C. Mode
- D. Range

4. Statistics, STD2 S1 2009 HSC 2 MC

⚠ RAP Data (vs state mean) - Bottom 6%: School result (66%) was 17% below state average (83%)

The step graph shows the charges for a carpark.



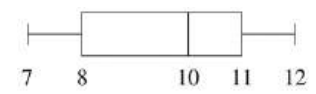
Maria enters the carpark at 10:10 am and exits at 1:30 pm.

How much will she pay in charges?

- (A) \$6
- (B) \$12
- (C) \$18
- (D) \$24

5. Statistics, STD2 S1 2004 HSC 12 MC

This box-and-whisker plot represents a set of scores.

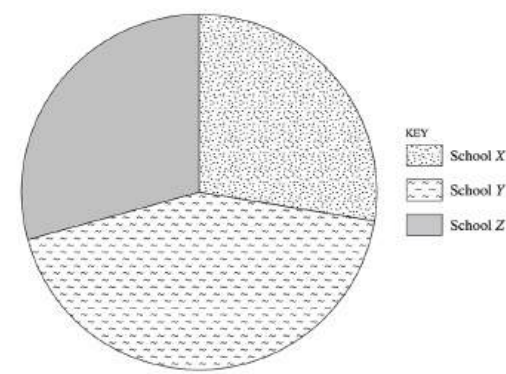


What is the interquartile range of this set of scores?

- A. 1
- B. 2
- C. 3
- D. 5

6. Statistics, STD2 S1 2004 HSC 8 MC

This sector graph shows the distribution of 116 prizes won by three schools: X, Y and Z.



How many prizes were won by School X?

- (A) 26
- (B) 32
- (C) 81
- (D) 99

7. Statistics, STD2 S5 2017 HSC 13 MC

The heights of Year 12 girls are normally distributed with a mean of 165 cm and a standard deviation of 5.5 cm.

What is the *z*-score for a height of 154 cm?

- A. -2
- B. -0.5
- C. 0.5
- D. 2

8. Statistics, STD2 S1 2018 HSC 1 MC

A set of scores has the following five-number summary.

- lower extreme = 2
- lower quartile = 5
- median = 6
- upper quartile = 8
- upper extreme = 9

What is the range?

- A. 2
- B. 3
- C. 6
- D. 7

9. Statistics, STD2 S1 SM-Bank 1 MC

A survey asked the following question for students born in Australia:

"Which State or Territory were you born in?"

How would the responses be classified?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, discrete
- D. Numerical, continuous

10. Statistics, STD2 S1 SM-Bank 2 MC

A dataset has the following five-number summary.

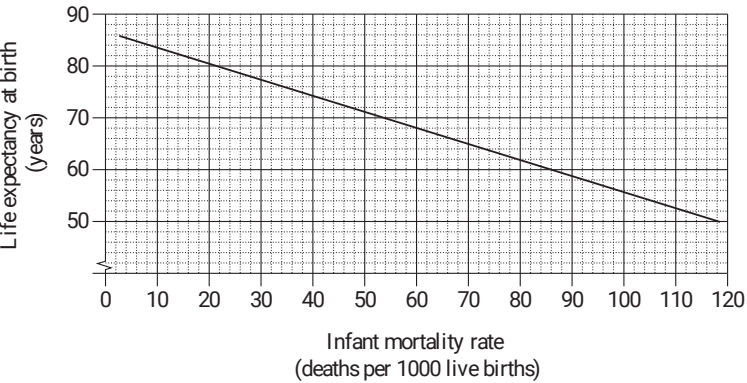
Minimum value	?
First quartile	10
Median	12
Third quartile	13
Maximum value	15

If the range of the dataset is 8, what is the minimum value of the dataset?

- A. 2
- B. 3
- C. 4
- D. 7

11. Algebra, STD2 A2 2017 HSC 3 MC

The graph shows the relationship between infant mortality rate (deaths per 1000 live births) and life expectancy at birth (in years) for different countries.

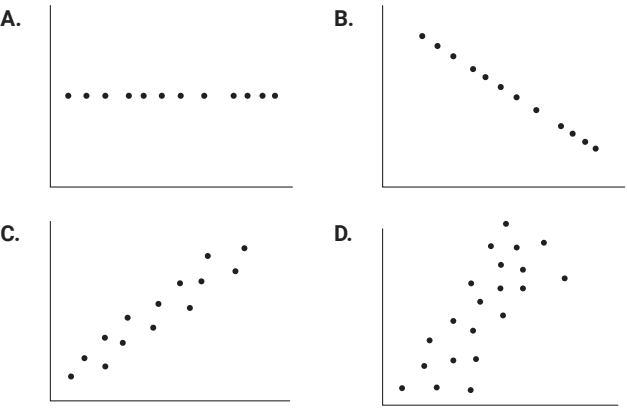


What is the life expectancy at birth in a country which has an infant mortality rate of 60?

- A. 68 years
- B. 69 years
- C. 86 years
- D. 88 years

12. Statistics, STD2 S4 2017 HSC 12 MC

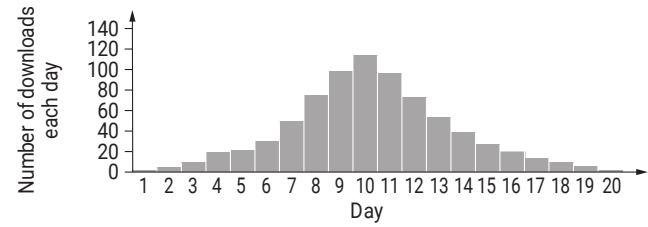
Which of the data sets graphed below has the largest positive correlation coefficient value?



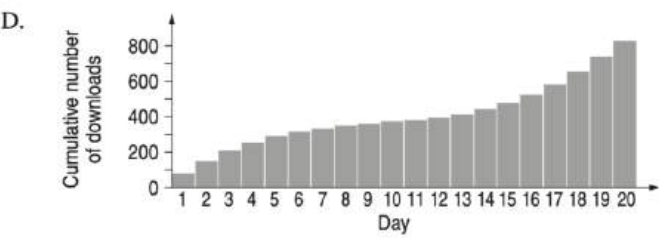
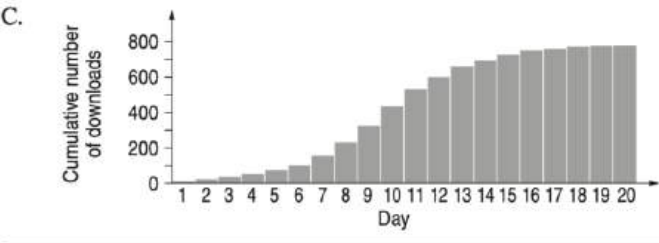
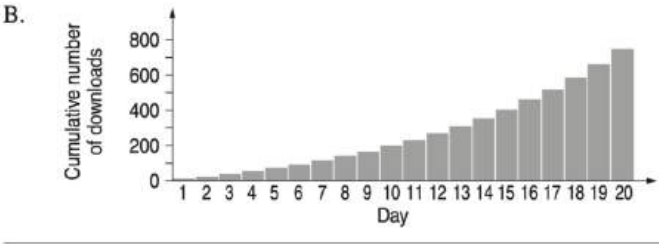
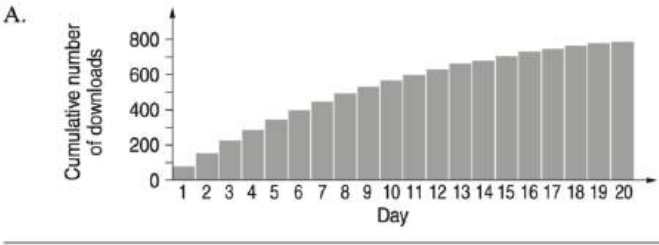
13. Statistics, 2ADV S2 2021 HSC 4 MC

⚡ *RAP Data (vs state mean) - Bottom 17%: School result (78%) was 1% below state average (79%)*

The number of downloads of a song on each of twenty consecutive days is shown in the following graph.

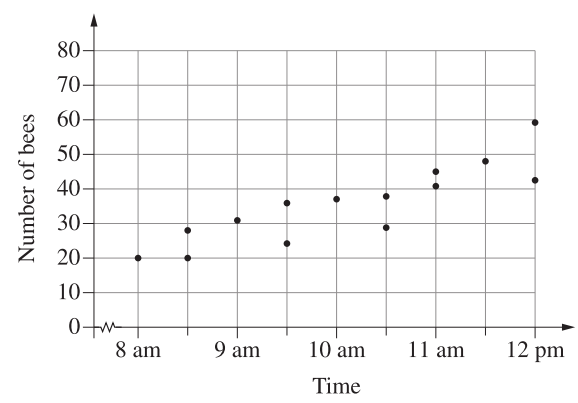


Which of the following graphs best shows the cumulative number of downloads up to and including each day?



14. Statistics, 2ADV S2 2023 HSC 1 MC

The number of bees leaving a hive was observed and recorded over 14 days at different times of the day.



Which Pearson's correlation coefficient best describes the observations?

- A. -0.8
- B. -0.2
- C. 0.2
- D. 0.8

15. Statistics, 2ADV S2 SM-Bank 5 MC

The stem plot below shows the height, in centimetres, of 20 players in a junior football team.

key: 14|2 = 142 cm $n = 20$

14		2	2	4	7	8	8	9	
15		0	0	1	2	5	5	6	8
16		0	1	1	2				
17		9							

A player with a height of 179 cm is considered an outlier because 179 cm is greater than

- A. 162 cm
- B. 169 cm
- C. 173 cm
- D. 175.5 cm

16. Statistics, 2ADV S3 SM-Bank 8 MC

The heights of females living in a small country town are normally distributed:

- 16% of the females are more than 160 cm tall.
- 2.5% of the females are less than 115 cm tall.

The mean and the standard deviation of this female population, in centimetres, are closest to

- A. mean = 135 standard deviation = 15
- B. mean = 135 standard deviation = 25
- C. mean = 145 standard deviation = 15
- D. mean = 145 standard deviation = 20

17. Statistics, STD2 S5 SM-Bank 4 MC

The length of a type of ant is approximately normally distributed with a mean of 4.8 mm and a standard deviation of 1.2 mm.

A standardised ant length of $z = -0.5$ corresponds to an actual ant length of

- A. 2.4 mm
- B. 3.6 mm
- C. 4.2 mm
- D. 5.4 mm

18. Statistics, STD2 S5 SM-Bank 1 MC

The head circumference (in cm) of a population of infant boys is normally distributed with a mean of 49.5 cm and a standard deviation of 1.5 cm.

Four hundred of these boys are selected at random and each boy's head circumference is measured.

The number of these boys with a head circumference of less than 48.0 cm is closest to

- (A) 3
- (B) 10
- (C) 64
- (D) 272

19. Statistics, STD2 S1 2004 HSC 6-7 MC

Use the set of scores 1, 3, 3, 3, 4, 5, 7, 7, 12 to answer Questions 6 and 7.

Question 6

What is the range of the set of scores?

- A. 6
- B. 9
- C. 11
- D. 12

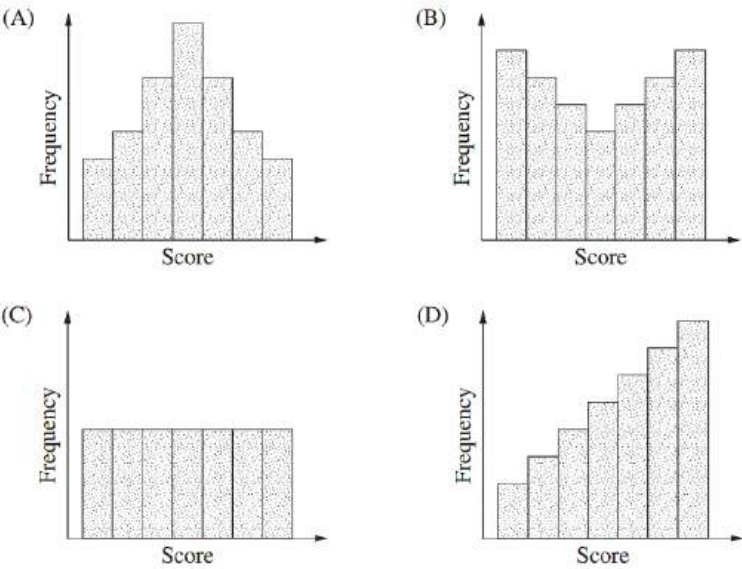
Question 7

What are the median and the mode of the set of scores?

- A. Median 3, mode 5
- B. Median 3, mode 3
- C. Median 4, mode 5
- D. Median 4, mode 3

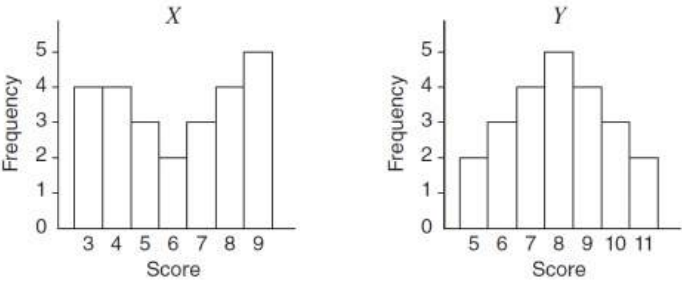
20. Statistics, STD2 S5 2010 HSC 4 MC

Which of the following frequency histograms shows data that could be normally distributed?



21. Statistics, STD2 S1 2011 HSC 11 MC

The sets of data, **X** and **Y**, are displayed in the histograms.



Which of these statements is true?

- (A) **X** has a larger mode and **Y** has a larger range.
- (B) **X** has a larger mode and the ranges are the same.
- (C) The modes are the same and **Y** has a larger range.
- (D) The modes are the same and the ranges are the same.

22. Statistics, STD2 S1 2011 HSC 14 MC

A data set of nine scores has a median of 7.

The scores 6, 6, 12 and 17 are added to this data set.

What is the median of the data set now?

- A. 6
- B. 7
- C. 8
- D. 9

23. Statistics, STD2 S1 2012 HSC 1 MC

A set of 15 scores is displayed in a stem-and-leaf plot.

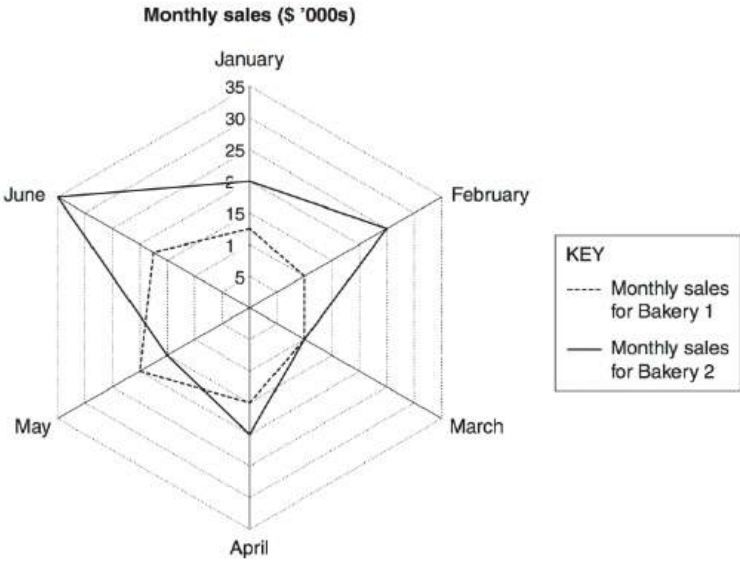


What is the median of these scores?

- (A) 7
- (B) 8
- (C) 77
- (D) 78

24. Statistics, STD2 S1 2012 HSC 7 MC

The Pi Company has two bakeries. The radar chart displays the monthly sales for the bakeries.

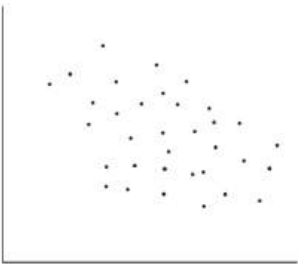


What was the difference in sales in June between the two bakeries?

- (A) \$7.50
- (B) \$17.50
- (C) \$7500
- (D) \$17 500

25. Statistics, STD2 S4 2016 HSC 3 MC

The graph shows a scatterplot for a set of data.



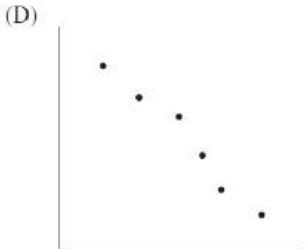
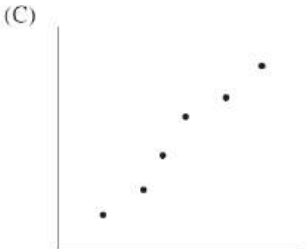
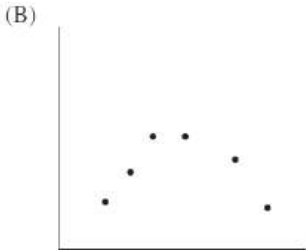
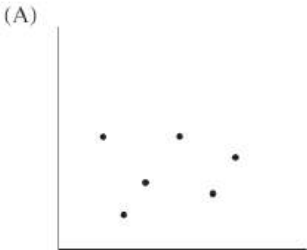
Which of the following is the best approximation for the correlation coefficient of this set of data?

- (A) -1
- (B) -0.3
- (C) 0.3
- (D) 1

26. Statistics, STD2 S4 2011 HSC 8 MC

⚠ RAP Data (vs state mean) - Bottom 8%: School result (54%) was 15% below state average (69%)

In which graph would the data have a correlation coefficient closest to -0.9 ?



27. Statistics, STD2 S1 2011 HSC 17 MC

⚠ RAP Data (vs state mean) - Bottom 4%: School result (42%) was 19% below state average (61%)

The heights of the players in a basketball team were recorded as 1.8 m, 1.83 m, 1.84 m, 1.86 m and 1.92 m. When a sixth player joined the team, the average height of the players increased by 1 centimetre.

What was the height of the sixth player?

- (A) 1.85 m
- (B) 1.86 m
- (C) 1.91 m
- (D) 1.93 m

28. Statistics, STD2 S1 2013 HSC 14 MC

The July sales prices for properties in a suburb were:

\$552 000, \$595 000, \$607 000, \$607 000, \$682 000, and \$685 000.

On 1 August, another property in the same suburb was sold for over one million dollars.

If the property had been sold in July, what effect would it have had on the mean and median sale prices for July?

- (A) Both the mean and median would have changed.
- (B) Neither the mean nor the median would have changed.
- (C) The mean would have changed and the median would have stayed the same.
- (D) The mean would have stayed the same and the median would have changed.

29. Statistics, STD2 S1 2013 HSC 15 MC

The frequency histogram shows the number of goals scored by a football team in each game in a season.

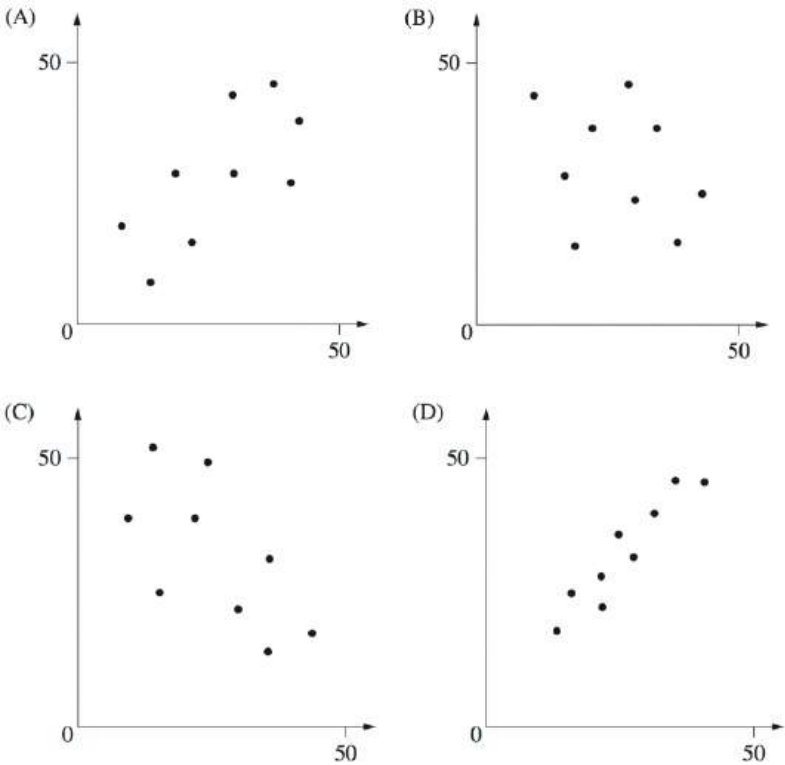


What is the mean number of goals scored per game by this team?

- (A) 4
- (B) 4.5
- (C) 5
- (D) 5.5

30. Statistics, STD2 S4 2013 HSC 2 MC

Which graph best shows data with a correlation closest to 0.3?



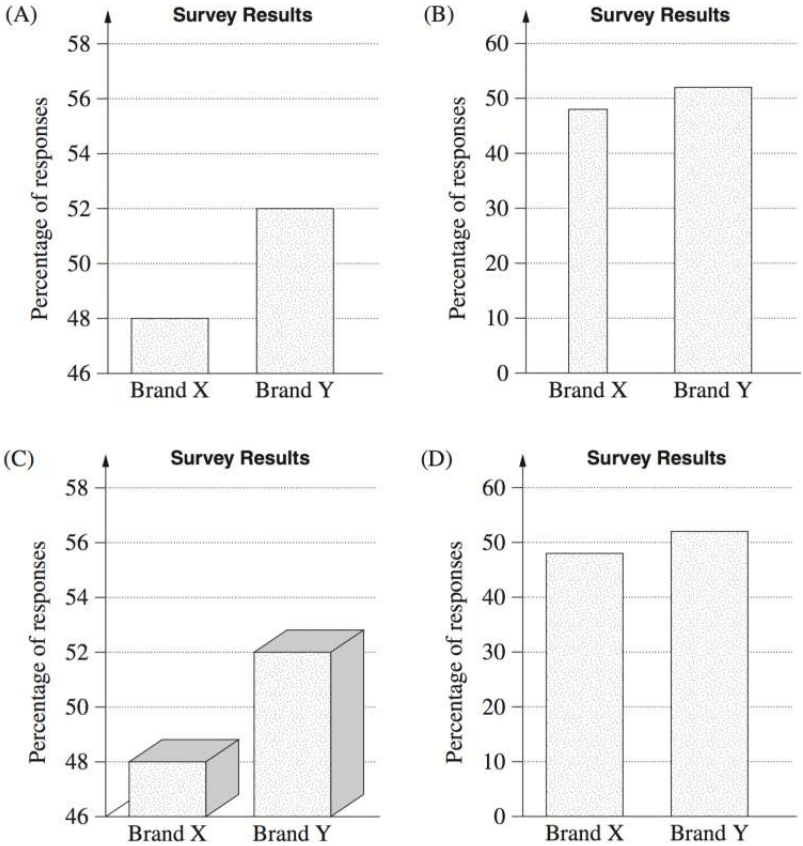
31. Statistics, STD2 S1 2013 HSC 6 MC

A survey was conducted where people were asked which of two brands of smartphones they preferred. The results were:

- 48% preferred Brand X
- 52% preferred Brand Y

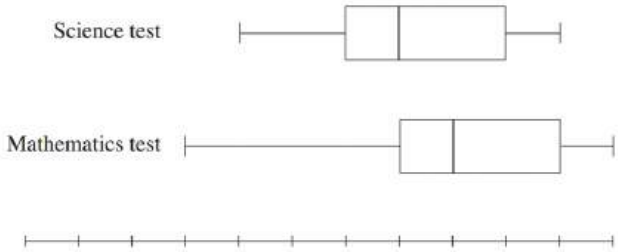
A graph displaying the data is to be included in a magazine article. The editor of the magazine wishes to ensure that the graph is not misleading in any way.

Which graph should the editor choose to include in the article?



32. Statistics, STD2 S1 2008 HSC 10 MC

The marks for a Science test and a Mathematics test are presented in box-and-whisker plots.



Which measure must be the same for both tests?

- A. Mean
- B. Range
- C. Median
- D. Interquartile range

33. Statistics, STD2 S1 2008 HSC 13 MC

The height of each student in a class was measured and it was found that the mean height was 160 cm.

Two students were absent. When their heights were included in the data for the class, the mean height did not change.

Which of the following heights are possible for the two absent students?

- (A) 155 cm and 162 cm
- (B) 152 cm and 167 cm
- (C) 149 cm and 171 cm
- (D) 143 cm and 178 cm

34. Statistics, STD2 S1 2008 HSC 8 MC

What is the median of the following set of scores?

<i>Score</i>	<i>Frequency</i>
12	13
14	6
16	2
18	12
Total	33

- (A) 12
- (B) 13
- (C) 14
- (D) 15

35. Statistics, STD2 S1 2006 HSC 12 MC

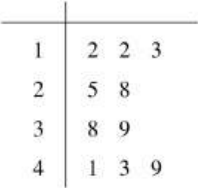
The mean of a set of 5 scores is 62.

What is the new mean of the set of scores after a score of 14 is added?

- (A) 38
- (B) 54
- (C) 62
- (D) 76

36. Statistics, STD2 S1 2006 HSC 4 MC

A set of scores is displayed in a stem-and-leaf plot.



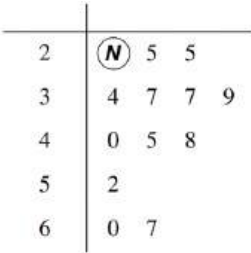
What is the median of this set of scores?

- (A) 28
- (B) 30
- (C) 33
- (D) 47

37. Statistics, STD2 S1 2008 HSC 3 MC

The stem-and-leaf plot represents the daily sales of soft drink from a vending machine.

If the range of sales is 43, what is the value of **N** ?



- (A) 4
- (B) 5
- (C) 24
- (D) 25

38. Statistics, STD2 S1 2015 HSC 4 MC

On a school report, a student's record of completing homework is graded using the following codes.

- C = consistently
- U = usually
- S = sometimes
- R = rarely
- N = never

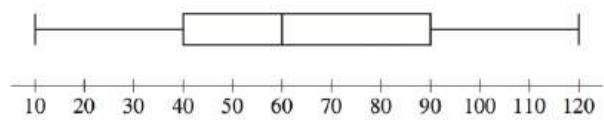
What type of data is this?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, continuous
- D. Numerical, discrete

39. Statistics, STD2 S1 2015 HSC 6 MC

⚡ RAP Data (vs state mean) - Bottom 16%: School result (53%) was 9% below state average (62%)

The times, in minutes, that a large group of students spend on exercise per day are presented in the box-and-whisker plot.



What percentage of these students spend between 40 minutes and 60 minutes per day on exercise?

- A. 17%
- B. 20%
- C. 25%
- D. 50%

40. Statistics, STD2 S5 2016 HSC 13 MC

⚡ RAP Data (vs state mean) - Bottom 10%: School result (45%) was 13% below state average (58%)

The speed limit outside a school is 40 km/h. Year 11 students measured the speed of passing vehicles over a period of time. They found the set of data to be normally distributed with a mean speed of 36 km/h and a standard deviation of 2 km/h.

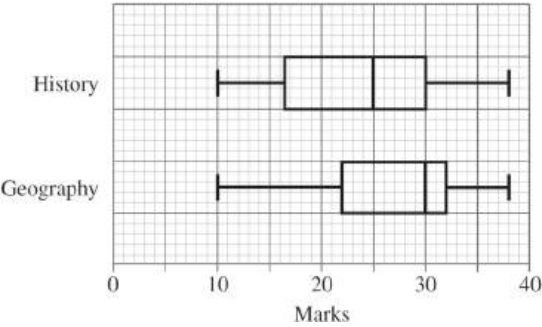
What percentage of the vehicles passed the school at a speed greater than 40 km/h?

- (A) 2.5%
- (B) 5%
- (C) 47.5%
- (D) 95%

41. Statistics, STD2 S1 2016 HSC 22 MC

⚠ RAP Data (vs state mean) - Bottom 6%: School result (45%) was 17% below state average (62%)

The box-and-whisker plots show the results of a History test and a Geography test.



In History, 112 students completed the test. The number of students who scored above 30 marks was the same for the History test and the Geography test.

How many students completed the Geography test?

- A. 8
- B. 50
- C. 56
- D. 112

42. Statistics, STD2 S1 2018 HSC 3 MC

A survey asked the following question.

'How many brothers do you have?'

How would the responses be classified?

- A. Categorical, ordinal
- B. Categorical, nominal
- C. Numerical, discrete
- D. Numerical, continuous

43. Statistics, STD2 S1 2018 HSC 11 MC

A set of data is summarised in this frequency distribution table.

<i>Score</i>	<i>Frequency</i>
3	1
4	2
5	6
6	7
7	9
8	5
Total = 30	

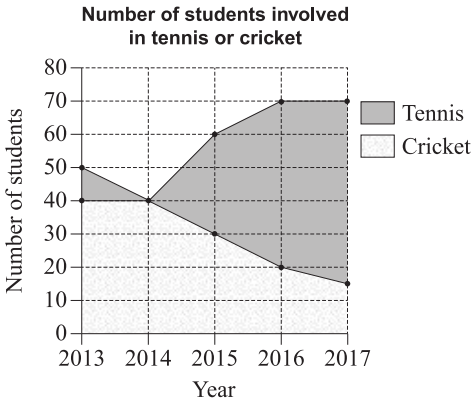
Which of the following is true about the data?

- A. Mode = 7, median = 5.5
- B. Mode = 7, median = 6
- C. Mode = 9, median = 5.5
- D. Mode = 9, median = 6

44. Statistics, STD2 S1 2018 HSC 17 MC

i *RAP Data (vs state mean) - Bottom 25%: School result (51%) was 5% below state average (56%)*

The area chart shows the number of students involved in tennis or cricket at a school over a number of years.

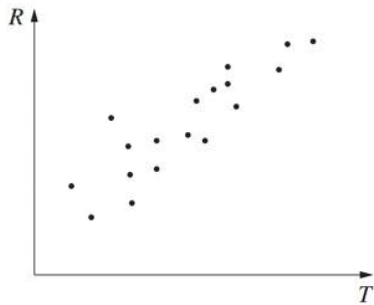


In which year was the number of students involved in tennis equal to the number of students involved in cricket?

- A. 2013
- B. 2014
- C. 2015
- D. 2016

45. Statistics, STD2 S4 2008 HSC 12 MC

A scatterplot is shown.



Which of the following best describes the correlation between ***R*** and ***T***?

- A. Positive
- B. Negative
- C. Positively skewed
- D. Negatively skewed

46. Statistics, STD2 S4 2007 HSC 9 MC

Which of the following would be most likely to have a positive correlation?

- A. The population of a town and the number of schools in that town
- B. The price of petrol per litre and the number of litres of petrol sold
- C. The hours training for a marathon and the time taken to complete the marathon
- D. The number of dogs per household and the number of televisions per household

47. Statistics, 2ADV S3 2020 HSC 3 MC

John recently did a class test in each of three subjects. The class scores on each test were normally distributed.

The table shows the subjects and John's scores as well as the mean and standard deviation of the class scores on each test.

<i>Subject</i>	<i>John's score</i>	<i>Mean</i>	<i>Standard deviation</i>
French	82	70	8
Commerce	80	65	5
Music	74	50	12

Relative to the rest of class, which row of the table below shows John's strongest subject and his weakest subject?

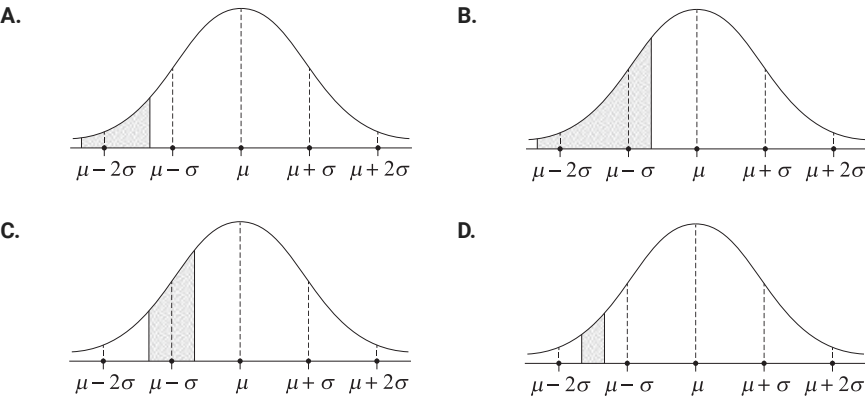
	<i>Strongest subject</i>	<i>Weakest subject</i>
A.	Commerce	French
B.	French	Music
C.	Music	French
D.	Commerce	Music

48. Statistics, 2ADV S3 2020 HSC 9 MC

⚡ RAP Data (vs state mean) - Bottom 11%: School result (51%) was 4% below state average (55%)

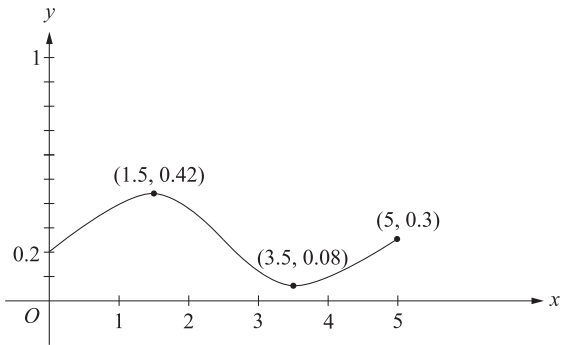
Suppose the weight of melons is normally distributed with a mean of μ and a standard deviation of σ . A melon has a weight below the lower quartile of the distribution but NOT in the bottom 10% of the distribution.

Which of the following most accurately represents the region in which the weight of this melon lies?



49. Statistics, 2ADV S3 EQ-Bank 6 MC

A continuous probability density function graph is drawn below.



Which of the following is the mode?

- A. 0.42
- B. 0.08
- C. 1.5
- D. 5

50. Statistics, 2ADV S3 SM-Bank 5 MC

The function $f(x)$ is a probability density function of a continuous random variable with the rule

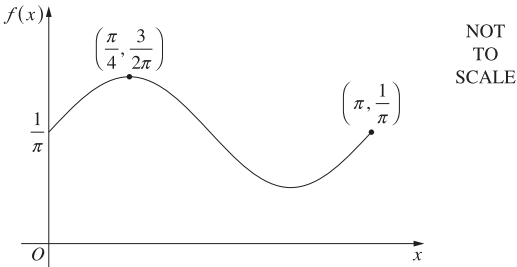
$$f(x) = \begin{cases} ae^x & 0 \leq x \leq 1 \\ ae & 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The value of a is

- A. 1
- B. $\frac{1}{e}$
- C. $\frac{1}{2e}$
- D. $\frac{1}{2e - 1}$

51. Statistics, 2ADV S3 2022 HSC 7 MC

Consider the following graph of a probability density function $f(x)$.



What is the value of the mode?

- A. $\frac{1}{\pi}$
- B. $\frac{3}{2\pi}$
- C. $\frac{\pi}{4}$
- D. π

52. Statistics, STD2 S5 2021 HSC 8 MC

On a test, Zac's mark corresponded to a z -score of 2. The test scores had a mean of 63 and a standard deviation of 8.

What was Zac's actual mark on the test?

- A. 65
- B. 67
- C. 73
- D. 79

53. Statistics, STD2 S5 SM-Bank 3 MC

The time, in hours, that each student spent sleeping on a school night was recorded for **1550** secondary-school students. The distribution of these times was found to be approximately normal with a mean of 7.4 hours and a standard deviation of 0.7 hours.

How many students would you expect to spend more than 8.1 hours sleeping on a school night?

You may assume for normally distributed data that:

- **68%** of scores have *z*-scores between **-1** and **1**
- **95%** of scores have *z*-scores between **-2** and **2**
- **99.7%** of scores have *z*-scores between **-3** and **3**.

- A. 16
- B. 248
- C. 1302
- D. 1510

54. Statistics, STD2 S5 SM-Bank 2 MC

The pulse rates of a large group of 18-year-old students are approximately normally distributed with a mean of 75 beats/minute and a standard deviation of 11 beats/minute.

The percentage of 18-year-old students with pulse rates less than 53 beats/minute or greater than 86 beats/minute is closest to

- (A) 2.5%
- (B) 5%
- (C) 16%
- (D) 18.5%

55. Statistics, STD2 S4 2012 HSC 11 MC

- ⚠ *RAP Data (vs state mean) - Bottom 4%: School result (23%) was 20% below state average (43%)*
- ⚠ *RAP Data (raw) - School result was 23%*

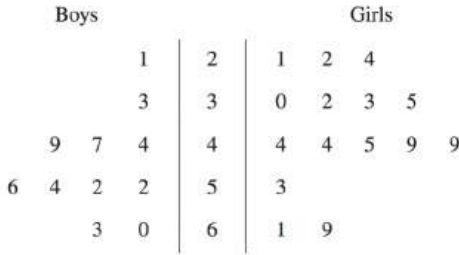
Which of the following relationships would most likely show a negative correlation?

- A. The population of a town and the number of hospitals in that town.
- B. The hours spent training for a race and the time taken to complete the race.
- C. The price per litre of petrol and the number of people riding bicycles to work.
- D. The number of pets per household and the number of computers per household.

56. Statistics, STD2 S1 2010 HSC 16 MC

- ⓘ *RAP Data (vs state mean) - Bottom 25%: School result (30%) was 5% below state average (35%)*

This back-to-back stem-and-leaf plot displays the test results for a class of 26 students.



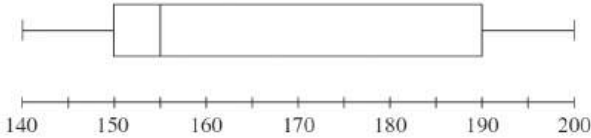
What is the median test result for the class?

- (A) 44
- (B) 46
- (C) 48
- (D) 49

57. Statistics, STD2 S1 2011 HSC 7 MC

- ⚠ *RAP Data (vs state mean) - Bottom 3%: School result (24%) was 23% below state average (47%)*
- ⚠ *RAP Data (raw) - School result was 24%*

A set of data is displayed in this box-and-whisker plot.

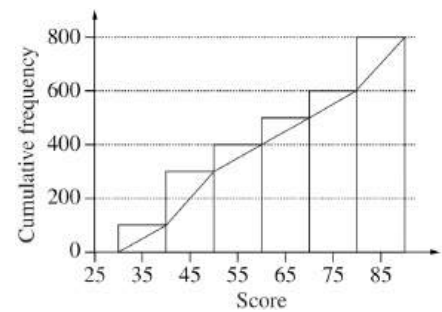


Which of the following best describes this set of data?

- A. Symmetrical
- B. Positively skewed
- C. Negatively skewed
- D. Normally distributed

58. Statistics, STD2 S1 2005 HSC 9 MC

A set of data is represented by the cumulative frequency histogram and ogive.



What is the best approximation for the interquartile range for this set of data?

- (A) 25
- (B) 30
- (C) 35
- (D) 40

59. Statistics, STD2 S5 2006 HSC 17 MC

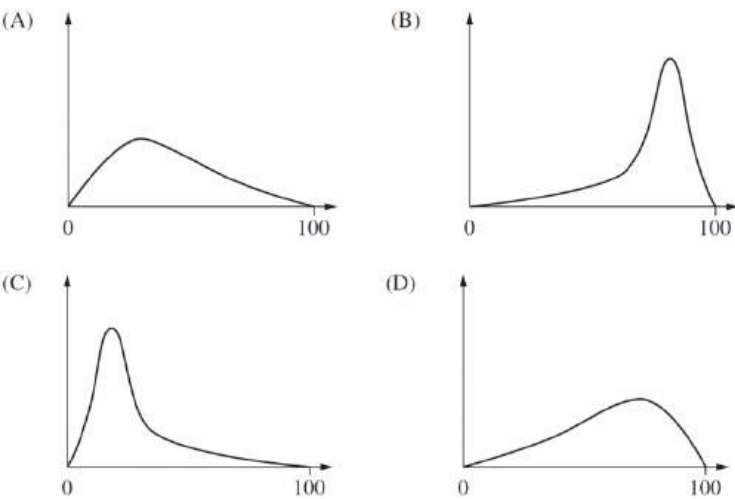
In a normally distributed set of scores, the mean is 23 and the standard deviation is 5.

Approximately what percentage of the scores will lie between 18 and 33?

- (A) 34%
- (B) 47.5%
- (C) 68%
- (D) 81.5%

60. Statistics, STD2 S1 2006 HSC 8 MC

Which of these graphs best represents positively skewed data with the smaller standard deviation?



61. Statistics, STD2 S1 2007 HSC 21 MC

This set of data is arranged in order from smallest to largest.

5, 6, 11, x , 13, 18, 25

The range is six less than twice the value of x .

Which one of the following is true?

- (A) The median is 12 and the interquartile range is 7.
- (B) The median is 12 and the interquartile range is 12.
- (C) The median is 13 and the interquartile range is 7.
- (D) The median is 13 and the interquartile range is 12.

62. Statistics, STD2 S1 2014 HSC 14 MC

Twenty Year 12 students were surveyed. These students were asked how many hours of sport they play per week, to the nearest hour.

The results are shown in the frequency table.

<i>Hours per week</i>	<i>Frequency</i>
0–2	5
3–5	10
6–8	3
9–11	2

What is the mean number of hours of sport played by the students per week?

- (A) 3.3
- (B) 4.3
- (C) 5.0
- (D) 5.3

63. Statistics, STD2 S5 2014 HSC 24 MC

The weights of 10 000 newborn babies in NSW are normally distributed. These weights have a mean of 3.1 kg and a standard deviation of 0.35 kg.

How many of these newborn babies have a weight between 2.75 kg and 4.15 kg?

- (A) 4985
- (B) 6570
- (C) 8370
- (D) 8385

64. Statistics, STD2 S5 2015 HSC 20 MC

A machine produces cylindrical pipes. The mean of the diameters of the pipes is 8 cm and the standard deviation is 0.04 cm.

Assuming a normal distribution, what percentage of cylindrical pipes produced will have a diameter less than 7.96 cm?

- (A) 16%
- (B) 32%
- (C) 34%
- (D) 68%

65. Statistics, STD2 S1 2016 HSC 19 MC

A soccer referee wrote down the number of goals scored in 9 different games during the season.

2, 3, 3, 3, 5, 5, 8, 9, ...

The last number has been omitted. The range of the data is 10.

What is the five-number summary for this data set?

- A. 2, 3, 5, 8.5, 12
- B. 2, 3, 5, 8.5, 10
- C. 2, 3, 5, 8, 12
- D. 2, 3, 5, 8, 10

66. Statistics, STD2 S1 2016 HSC 21 MC

A grouped data frequency table is shown.

<i>Class interval</i>	<i>Frequency</i>
1–5	3
6–10	6
11–15	8
16–20	9

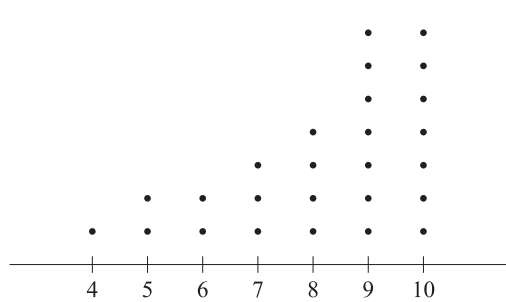
What is the mean for this set of data?

- (A) 6.5
- (B) 10.5
- (C) 11.9
- (D) 12.4

67. Statistics, STD2 S1 2018 HSC 6 MC

⚡ RAP Data (vs state mean) - Bottom 20%: School result (36%) was 7% below state average (43%)

A set of data is displayed in this dot plot.



Which of the following best describes this set of data?

- A. Symmetrical
- B. Positively skewed
- C. Negatively skewed
- D. Normally distributed

68. Statistics, STD2 S5 2018 HSC 23 MC

i RAP Data (vs state mean) - Bottom 25%: School result (41%) was 6% below state average (47%)

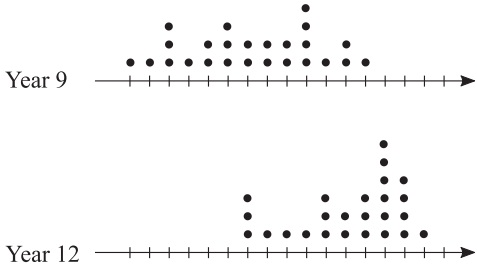
A set of data is normally distributed with a mean of 48 and a standard deviation of 3.

Approximately what percentage of the scores lies between 39 and 45?

- A. 15.85%
- B. 31.7%
- C. 47.5%
- D. 49.85%

69. Statistics, STD2 S1 SM-Bank 2 MC

The dot plots show the height of students in Year 9 and Year 12 in a school. They are drawn on the same scale.



Which statement about the change in heights when comparing Y9 to Y12 is correct?

- A. The mean increased and the standard deviation decreased.
- B. The mean decreased and the standard deviation decreased.
- C. The mean increased and the standard deviation increased.
- D. The mean decreased and the standard deviation increased.

70. Statistics, STD2 S1 2015 HSC 19 MC

The table shows the life expectancy (expected remaining years of life) for females at selected ages in the given periods of time.

Period of time	Life expectancy for females			
	at age 0	at age 25	at age 45	at age 65
1965–1967	74.2	51.2	32.3	15.7
1975–1977	76.6	53.1	34.0	17.1
1985–1987	79.2	55.4	36.1	18.6
1995–1997	81.3	57.1	37.7	19.8

In 1975, a 45-year-old female used the information in the table to calculate the age to which she was expected to live. Twenty years later she recalculated the age to which she was expected to live.

What is the difference between the two ages she calculated?

- (A) 2.7 years
- (B) 3.1 years
- (C) 3.7 years
- (D) 5.8 years

71. Statistics, 2ADV S3 SM-Bank 6 MC

The continuous random variable, $\mathbf{X}$, has a probability density function given by

$$f(x) = \begin{cases} \frac{1}{4}\cos\left(\frac{x}{2}\right) & 3\pi \leq x \leq 5\pi \\ 0 & \text{elsewhere} \end{cases}$$

The value of $\mathbf{a}$ such that $P(X < a) = \frac{\sqrt{3} + 2}{4}$ is

- A. $\frac{19\pi}{6}$
- B. $\frac{14\pi}{3}$
- C. $\frac{10\pi}{3}$
- D. $\frac{29\pi}{6}$

72. Statistics, 2ADV S3 SM-Bank 7 MC

A probability density function $\mathbf{f(x)}$ is given by

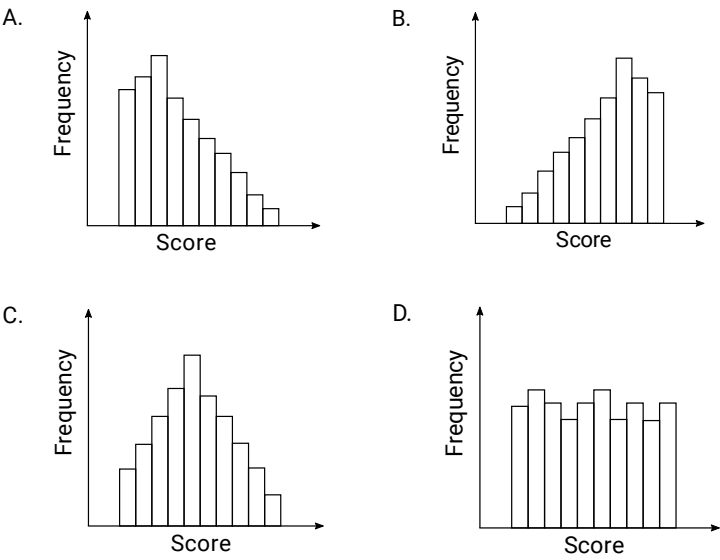
$$f(x) = \begin{cases} \frac{1}{12}(8x - x^3) & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

The median $\mathbf{m}$ of this function satisfies the equation

- A. $-m^4 + 16m^2 - 6 = 0$
- B. $m^4 - 16m^2 = 0$
- C. $m^4 - 16m^2 + 24 = 0.5$
- D. $m^4 - 16m^2 + 24 = 0$

73. Statistics, STD2 S1 2020 HSC 7 MC

Which histogram best represents a dataset that is positively skewed?



74. Statistics, STD2 S1 2009 HSC 21 MC

The mean of a set of ten scores is 14. Another two scores are included and the new mean is 16.

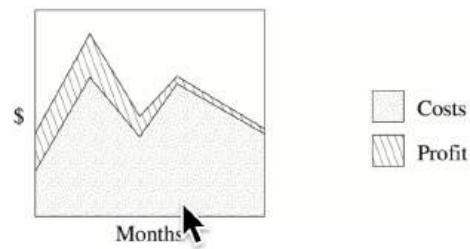
What is the mean of the two additional scores?

- (A) 4
- (B) 16
- (C) 18
- (D) 26

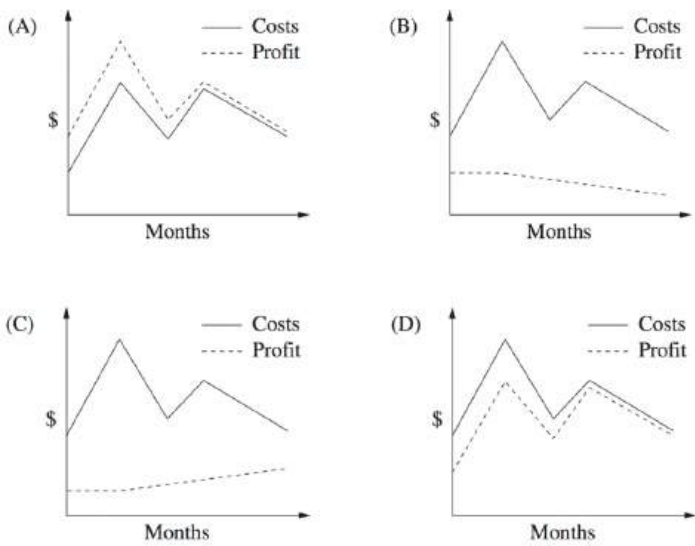
75. Statistics, STD2 S1 2010 HSC 21 MC

- ⚡ RAP Data (vs state mean) - Bottom 12%: School result (12%) was 12% below state average (24%)
- ⚡ RAP Data (raw) - School result was 12%

The area graph shows the cost and profits for a business over a period of time.



The information in the area graph is then displayed as a line graph.
Which of the following line graphs best displays the data from the area graph?



76. Statistics, STD2 S5 2013 HSC 20 MC

There are 60 000 students sitting a state-wide examination. If the results form a normal distribution, how many students would be expected to score a result between 1 and 2 standard deviations above the mean?

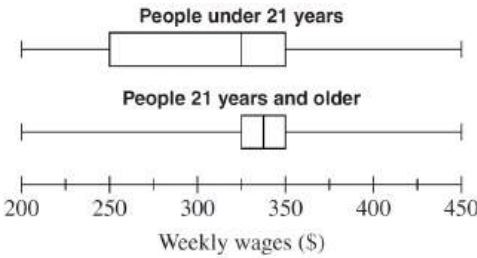
You may assume for normally distributed data that:

- 68% of scores have z -scores between -1 and 1
- 95% of scores have z -scores between -2 and 2
- 99.7% of scores have z -scores between -3 and 3 .

- (A) 8100
- (B) 16 200
- (C) 20 400
- (D) 28 500

77. Statistics, STD2 S1 2005 HSC 22 MC

Two groups of people were surveyed about their weekly wages. The results are shown in the box-and-whisker plots.

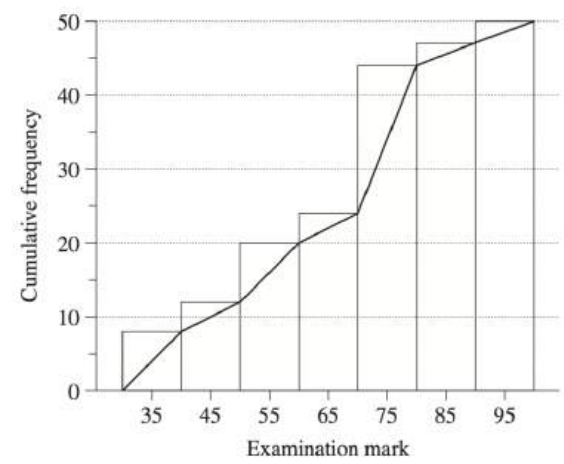


Which of the following statements is true for the people surveyed?

- A. The same percentage of people in each group earned more than \$325 per week.
- B. Approximately 75% of people under 21 years earned less than \$350 per week.
- C. Approximately 75% of people 21 years and older earned more than \$350 per week.
- D. Approximately 50% of people in each group earned between \$325 and \$350 per week.

78. Statistics, STD2 S1 2007 HSC 22 MC

A set of examination results is displayed in a cumulative frequency histogram and polygon (ogive).



Sanath knows that his examination mark is in the 4th decile.
Which of the following could have been Sanath's examination mark?

- (A) 37
- (B) 57
- (C) 67
- (D) 77

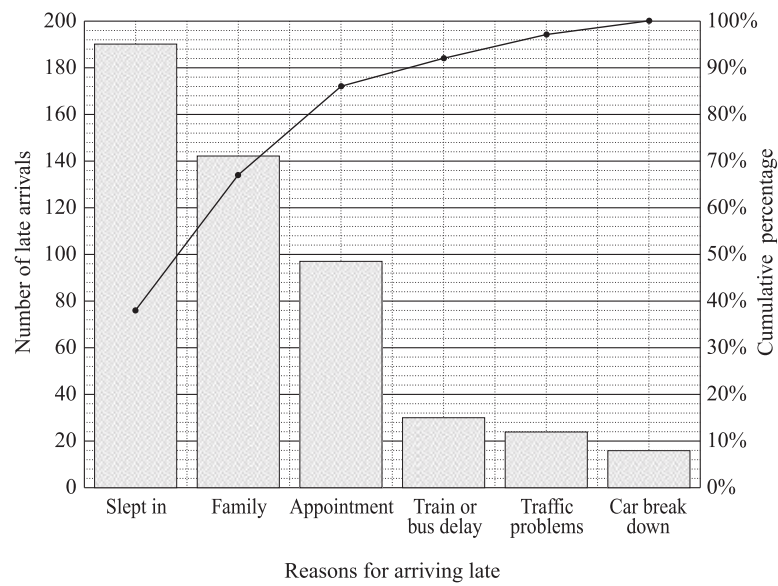
79. Statistics, STD2 S1 2016 HSC 7 MC

Which set of data is classified as categorical and nominal?

- A. blue, green, yellow
- B. small, medium, large
- C. 5.2 cm, 6 cm, 7.21 cm
- D. 4 people, 5 people, 9 people

80. Statistics, STD2 S1 2019 HSC 10 MC

A school collected data related to the reasons given by students for arriving late. The Pareto chart shows the data collected.



What percentage of students gave the reason 'Train or bus delay'?

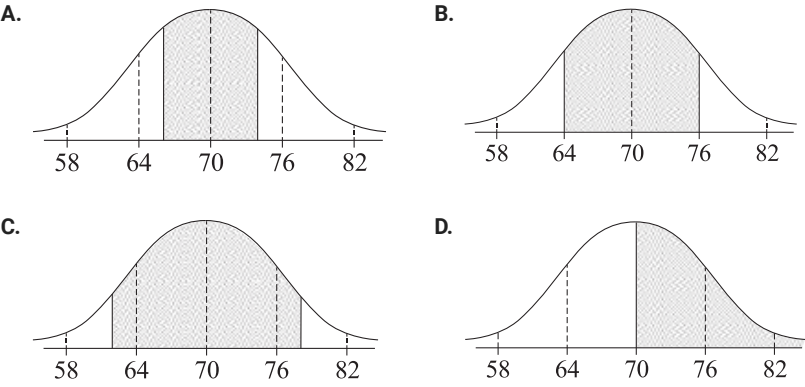
- A. 6%
- B. 15%
- C. 30%
- D. 92%

81. Statistics, STD2 S5 2019 HSC 15 MC

i RAP Data (raw) - School result was 15%

The scores on an examination are normally distributed with a mean of 70 and a standard deviation of 6. Michael received a score on the examination between the lower quartile and the upper quartile of the scores.

Which shaded region most accurately represents where Michael's score lies?



Worked Solutions

1. Statistics, 2ADV S2 2022 HSC 2 MC

Mean decreases.
Median remains 17.
 $\Rightarrow D$

◆ Mean mark 48%.

2. Statistics, STD2 S1 2010 HSC 1 MC

Range = High – Low
 $= 9 - 0$
 $= 9$
 $\Rightarrow C$

3. Statistics, STD2 S1 2009 HSC 3 MC

Eye colour is categorical data
 $\therefore$ **Only the mode can be found**
 $\Rightarrow C$

4. Statistics, STD2 S1 2009 HSC 2 MC

Time in carpark = 3h 20m
From graph, charge will be \$18
 $\Rightarrow C$

5. Statistics, STD2 S1 2004 HSC 12 MC

$$Q_1 = 8, Q_3 = 11$$

$$\begin{aligned} \text{IQR} &= Q_3 - Q_1 \\ &= 11 - 8 \\ &= 3 \end{aligned}$$

$\Rightarrow C$

6. Statistics, STD2 S1 2004 HSC 8 MC

$$\begin{aligned} \text{Centre angle of School X sector} \\ &= 100^\circ \text{ (by measurement)} \end{aligned}$$

$\therefore$ Prizes won by school X

$$\begin{aligned} &= \frac{100}{360} \times 116 \\ &= 32.22 \dots \end{aligned}$$

$\Rightarrow B$

7. Statistics, STD2 S5 2017 HSC 13 MC

$$\begin{aligned} z\text{-score} &= \frac{x - \mu}{\sigma} \\ &= \frac{154 - 165}{5.5} \\ &= -2 \end{aligned}$$

$\Rightarrow A$

8. Statistics, STD2 S1 2018 HSC 1 MC

$$\begin{aligned} \text{Range} &= \text{upper extreme} - \text{lower extreme} \\ &= 9 - 2 \\ &= 7 \end{aligned}$$

$\Rightarrow D$

9. Statistics, STD2 S1 SM-Bank 1 MC

The data is categorical (not numerical) since the name of a State is required.

This data cannot be ordered.

$\Rightarrow B$

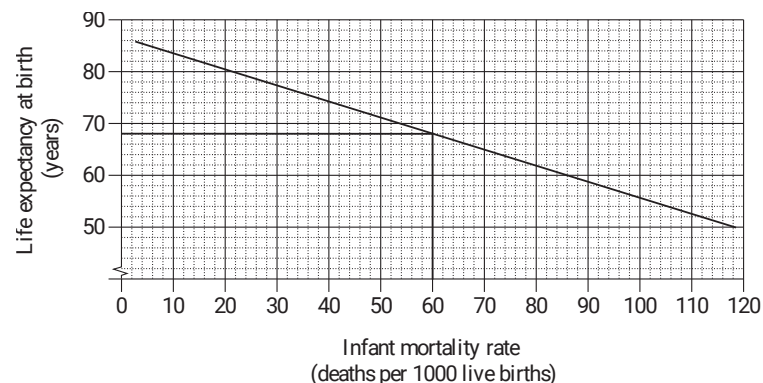
10. Statistics, STD2 S1 SM-Bank 2 MC

$$\begin{aligned} \text{Range} &= \text{Max} - \text{Min} \\ 8 &= 15 - \text{Min Value} \\ \therefore \text{Min} &= 15 - 8 \\ &= 7 \end{aligned}$$

$\Rightarrow D$

11. Algebra, STD2 A2 2017 HSC 3 MC

text(When infant mortality rate is 60, life expectancy at birth is 68 years (see below).



$\Rightarrow A$

12. Statistics, STD2 S4 2017 HSC 12 MC

Largest positive correlation occurs when both variables move in tandem. The tighter the linear relationship, the higher the correlation.

$\Rightarrow C$

(Note that B is negatively correlated)

13. Statistics, 2ADV S2 2021 HSC 4 MC

The gradient of the cumulative frequency histogram will increase gradually, be steepest at day 10 then decrease gradually.

$\Rightarrow C$

14. Statistics, 2ADV S2 2023 HSC 1 MC

Correlation is positive and strong.

Best option: $r = 0.8$ $\Rightarrow D$

NOTE: Inputting all data points into a calculator is unnecessary and time consuming here.

15. Statistics, 2ADV S2 SM-Bank 5 MC

$$Q_1 = \frac{148 + 148}{2} = 148$$

$$Q_3 = \frac{158 + 160}{2} = 159$$

$$IQR = 159 - 148 = 11$$

$$\begin{aligned}\text{Upper fence} &= Q_3 + 1.5 \times IQR \\ &= 159 + 1.5 \times 11 \\ &= 175.5\end{aligned}$$

 $\Rightarrow D$

16. Statistics, 2ADV S3 SM-Bank 8 MC

$$160 \rightarrow z\text{-score} = 1$$

$$115 \rightarrow z\text{-score} = -2$$

$$1 = \frac{160 - \mu}{\sigma}$$

$$\mu + \sigma = 160 \dots (1)$$

$$-2 = \frac{115 - \mu}{\sigma}$$

$$\mu - 2\sigma = 115 \dots (2)$$

$$(1) - (2)$$

$$3\sigma = 45$$

$$\sigma = 15$$

Substitute $\sigma = 15$ into (1)

$$\mu = 145$$

 $\Rightarrow C$

17. Statistics, STD2 S5 SM-Bank 4 MC

$$z = \frac{x - \mu}{\sigma}$$

$$-0.5 = \frac{x - 4.8}{1.2}$$

$$\begin{aligned}-0.6 &= x - 4.8 \\ x &= 4.2 \text{ mm}\end{aligned}$$

 $\Rightarrow C$

18. Statistics, STD2 S5 SM-Bank 1 MC

$$\mu = 49.5, \sigma = 1.5$$

$$\begin{aligned}z\text{-score}(49.5) &= \frac{x - \mu}{\sigma} \\ &= \frac{48.0 - 49.5}{1.5} \\ &= -1\end{aligned}$$

 $\therefore$ Number of boys with a head under 48.0 cm

$$= 16\% \times 400$$

$$= 64$$

 $\Rightarrow C$

19. Statistics, STD2 S1 2004 HSC 6-7 MC

Question 6

Range = High - Low

$$= 12 - 1$$

$$= 11$$

 $\Rightarrow C$

Question 7

9 scores

$$\therefore \text{Median} = \frac{9 + 1}{2}$$

$$= 5\text{th score}$$

$$= 4$$

Mode = 3

 $\Rightarrow D$

20. Statistics, STD2 S5 2010 HSC 4 MC

Normally distributed data have a frequency histogram graph that is shaped like a bell.
⇒ *A*

21. Statistics, STD2 S1 2011 HSC 11 MC

Mode of $X = 9$
Range of $X = 9 - 3 = 6$
Mode of $Y = 8$
Range of $Y = 11 - 5 = 6$
∴ X has a larger mode and ranges are the same
⇒ *B*

◆ Mean mark 47%

22. Statistics, STD2 S1 2011 HSC 14 MC

Since an even amount of scores are added below and above the existing median, it will not change.
⇒ *B*

23. Statistics, STD2 S1 2012 HSC 1 MC

15 scores ⇒ Median is 8th
∴ Median is 78
⇒ *D*

24. Statistics, STD2 S1 2012 HSC 7 MC

Bakery 1 Sales in June = \$17 500
Bakery 2 Sales in June = \$35 000
∴ Difference in Sales = 35 000 – 17 500
= 17 500
⇒ *D*

25. Statistics, STD2 S4 2016 HSC 3 MC

Correlation is negative and weak.
⇒ *B*

26. Statistics, STD2 S4 2011 HSC 8 MC

Data needs to show a strong negative correlation (i.e. top left to lower right.)
⇒ *D*

27. Statistics, STD2 S1 2011 HSC 17 MC

$$\begin{aligned}\text{Old Mean} &= (1.8 + 1.83 + 1.84 + 1.86 + 1.92) \div 5 \\ &= \frac{9.25}{5} \\ &= 1.85 \text{ m}\end{aligned}$$

Since the new mean = 1.86m (given)

New Mean = Height of all 6 players ÷ 6

$$\begin{aligned}\therefore 1.86 &= \frac{9.25 + h}{6} \quad (h = \text{height of new player}) \\ h &= (6 \times 1.86) - 9.25 \\ &= 1.91 \text{ m} \\ \Rightarrow C\end{aligned}$$

28. Statistics, STD2 S1 2013 HSC 14 MC

Mean increases because new house is sold above the existing average.

$$\begin{aligned}\text{Initial median} &= \frac{607\,000 + 607\,000}{2} = 607\,000 \\ \text{New median} &= 607\,000 \quad (4\text{th value in a list of } 7) \\ \Rightarrow C\end{aligned}$$

29. Statistics, STD2 S1 2013 HSC 15 MC

$$\begin{aligned}\text{Total number of goals scored} &= (3 \times 3) + (4 \times 7) + (5 \times 5) + (6 \times 1) + (7 \times 0) + (8 \times 4) \\ &= 9 + 28 + 25 + 6 + 0 + 32 \\ &= 100 \\ \text{Number of games} &= 3 + 7 + 5 + 1 + 4 = 20 \\ \therefore \text{Mean goals per game} &= \frac{100}{20} = 5 \\ \Rightarrow C\end{aligned}$$

30. Statistics, STD2 S4 2013 HSC 2 MC

A is correct since the data slopes bottom left to top right (i.e. it's positive).
D also slopes correctly but exhibits a higher correlation co-efficient.
⇒ *A*

31. Statistics, STD2 S1 2013 HSC 6 MC

D is the best as it starts at zero on the *y*-axis
and has the same column widths.

32. Statistics, STD2 S1 2008 HSC 10 MC

IQR = Upper Quartile – Lower Quartile

In both box plots, IQR = 3 intervals (against bottom scale)

$\Rightarrow D$

33. Statistics, STD2 S1 2008 HSC 13 MC

Since the mean doesn't change

$\Rightarrow$ **2 absent students must have a
mean height of 160 cm.**

Considering each option given,

$$(149 + 171) \div 2 = 160$$

$\Rightarrow C$

34. Statistics, STD2 S1 2008 HSC 8 MC

$$\begin{aligned}\text{Median} &= \frac{n+1}{2} \\ &= \frac{33+1}{2} \\ &= 17\text{th score}\end{aligned}$$

$\therefore$ **Median is 14**

$\Rightarrow C$

35. Statistics, STD2 S1 2006 HSC 12 MC

Mean of 5 scores = 62

$$\therefore \text{Total of 5 scores} = 62 \times 5 = 310$$

Add a score of 14

$$\text{Total of 6 scores} = 310 + 14 = 324$$

$$\begin{aligned}\therefore \text{New mean} &= \frac{324}{6} \\ &= 54\end{aligned}$$

$\Rightarrow B$

36. Statistics, STD2 S1 2006 HSC 4 MC

10 scores

$$\begin{aligned}\text{Median} &= \frac{5\text{th} + 6\text{th}}{2} \\ &= \frac{28 + 38}{2} \\ &= 33 \\ \Rightarrow C\end{aligned}$$

37. Statistics, STD2 S1 2008 HSC 3 MC

$$\text{Range} = \text{High} - \text{Low} = 43$$

$$\therefore 67 - \text{Low} = 43$$

$$\text{Low} = 24$$

$$\therefore N = 4$$

$\Rightarrow A$

38. Statistics, STD2 S1 2015 HSC 4 MC

**The data has been grouped into categories and
because each category can be ranked, it is ordinal.**

$\Rightarrow A$

39. Statistics, STD2 S1 2015 HSC 6 MC

$$Q_1 = 40, \text{ Median} = 60$$

$\therefore$ **% Students between 40 and 60**

$$= 50\% - 25\%$$

$$= 25\%$$

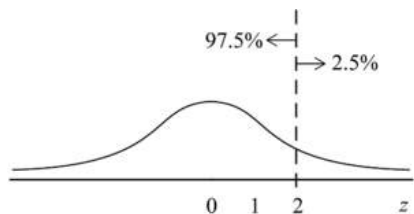
$\Rightarrow C$

40. Statistics, STD2 S5 2016 HSC 13 MC

$$z = \frac{x - \mu}{\sigma}$$

$$= \frac{40 - 36}{2}$$

$$= 2$$



⇒ A

41. Statistics, STD2 S1 2016 HSC 22 MC

In History ⇒ $Q_3 = 30$ marks

∴ Scoring over 30 = $25\% \times 112 = 28$ students

In Geography ⇒ Median = 30 marks

∴ Students completing Geography = $2 \times 28 = 56$ students

⇒ C

42. Statistics, STD2 S1 2018 HSC 3 MC

The number of brothers a person has is an exact whole number.

∴ Classification is numerical, discrete.

⇒ C

43. Statistics, STD2 S1 2018 HSC 11 MC

Mode = 7 (highest frequency of 9)

Median = average of 15th and 16th data points.

∴ Median = 6

⇒ B

44. Statistics, STD2 S1 2018 HSC 17 MC

Area Charts are cumulative,

Consider 2015,

Cricket players = 30

Tennis players = $60 - 30 = 30$

⇒ C

45. Statistics, STD2 S4 2008 HSC 12 MC

Correlation is positive.

NB. The skew does not directly relate to correlation.

⇒ A

46. Statistics, STD2 S4 2007 HSC 9 MC

Positive correlation means that as one variable increases, the other tends to increase also.

⇒ A

47. Statistics, 2ADV S3 2020 HSC 3 MC

Calculate the z -score of each subject:

$$z\text{-score (French)} = \frac{82 - 70}{8} = 1.5$$

$$z\text{-score (Commerce)} = \frac{80 - 65}{5} = 3.0$$

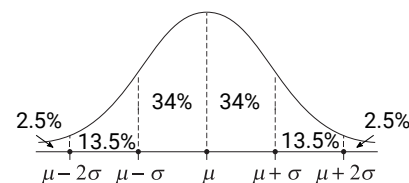
$$z\text{-score (Music)} = \frac{74 - 50}{12} = 2.0$$

∴ Commerce is strongest, French is weakest

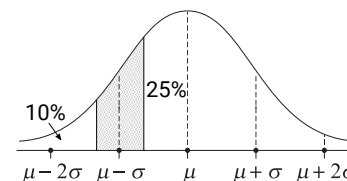
⇒ A

48. Statistics, 2ADV S3 2020 HSC 9 MC

Distributions using μ and σ :



Adjusting the above to identify the interval $10\% < \text{weight} < 25\%$



⇒ C

49. Statistics, 2ADV S3 EQ-Bank 6 MC

Mode is the most common x -value.

y -axis measures the probability.

Highest probability = 0.42

$$\therefore \text{Mode} = 1.5$$

$\Rightarrow C$

50. Statistics, 2ADV S3 SM-Bank 5 MC

Total area = 1

$$\int_0^1 ae^x dx + \int_1^2 ae dx = 1$$

$$[ae^x]_0^1 + [ae \cdot x]_1^2 = 1$$

$$[ae - a] + [2ae - ae] = 1$$

$$2ae - a = 1$$

$$a(2e - 1) = 1$$

$$\therefore a = \frac{1}{2e-1}$$

$\Rightarrow D$

51. Statistics, 2ADV S3 2022 HSC 7 MC

Mode $\rightarrow f(x)$ is a MAX

MAX occurs when $x = \frac{\pi}{4}$

$\Rightarrow C$

52. Statistics, STD2 S5 2021 HSC 8 MC

Method 1 (quicker)

$$\begin{aligned} \text{Actual mark} &= \mu + 2 \times \sigma \\ &= 63 + 2 \times 8 \\ &= 79 \end{aligned}$$

Method 2

$$z\text{-score} = \frac{x - \mu}{\sigma}$$

$$2 = \frac{x - 63}{8}$$

$$16 = x - 63$$

$$x = 79$$

$\Rightarrow D$

53. Statistics, STD2 S5 SM-Bank 3 MC

Need to find z -score of 8.1 hours

$$\begin{aligned} z\text{-score} &= \frac{x - \mu}{\sigma} \\ &= \frac{8.1 - 7.4}{0.7} \\ &= 1 \end{aligned}$$

68% students with $-1 < z\text{-score} < 1$

$\therefore$ 16% have $z\text{-score} > 1$

Students = 16% $\times$ 1550

$$= 248$$

$\Rightarrow B$

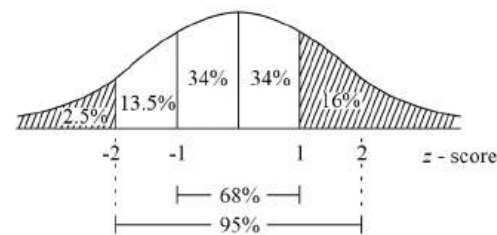
54. Statistics, STD2 S5 SM-Bank 2 MC

$$\mu = 75, \quad \sigma = 11$$

$$\begin{aligned} z\text{-score (53)} &= \frac{x - \mu}{\sigma} \\ &= \frac{53 - 75}{11} \\ &= -2 \end{aligned}$$

$$\begin{aligned} z\text{-score (86)} &= \frac{86 - 75}{11} \\ &= 1 \end{aligned}$$

COMMENT: Two applications of the 68-95-99.7% rule are required. A good strategy is to *first draw a normal curve* and shade the required areas.



From the diagram, the % of students that have a

z -score below -2 or above 1

$$= 2.5 + 16$$

$$= 18.5\%$$

$\Rightarrow D$

55. Statistics, STD2 S4 2012 HSC 11 MC

Increased hours training should reduce the time to complete a race.

$\Rightarrow B$

♦ Mean mark 43%.

56. Statistics, STD2 S1 2010 HSC 16 MC

26 results given in the data

$\Rightarrow$ Median is average of 13th and 14th

♦♦ Mean mark 35%

$$\begin{aligned}\therefore \text{Median} &= \frac{45 + 47}{2} \\ &= 46\end{aligned}$$

$\Rightarrow B$

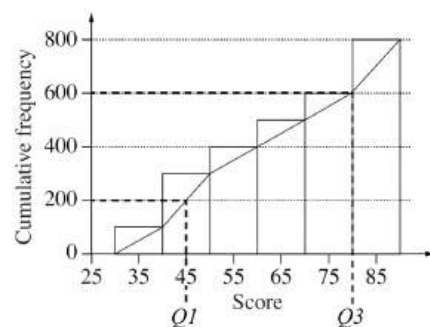
57. Statistics, STD2 S1 2011 HSC 7 MC

Since the median (155) is closer to the lower quartile (150) and range low (140) than the upper quartile (190) and range high (200), it is positively skewed.

♦ Mean mark 47%.

$\Rightarrow B$

58. Statistics, STD2 S1 2005 HSC 9 MC



$$\begin{aligned}IQR &= Q3 - Q1 \\ &= 80 - 45 \\ &= 35 \\ \Rightarrow C\end{aligned}$$

59. Statistics, STD2 S5 2006 HSC 17 MC

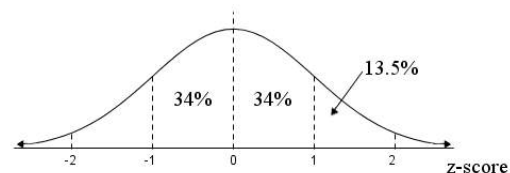
$$\mu = 23$$

$$\sigma = 5$$

$$\begin{aligned}z\text{-score}(18) &= \frac{x - \mu}{\sigma} \\ &= \frac{18 - 23}{5}\end{aligned}$$

$$= -1$$

$$\begin{aligned}z\text{-score}(33) &= \frac{33 - 23}{5} \\ &= 2\end{aligned}$$



$\therefore$ Percentage between -1 and 2

$$= 34 + 34 + 13.5$$

$$= 81.5\%$$

$\Rightarrow D$

60. Statistics, STD2 S1 2006 HSC 8 MC

By elimination

Positive skew when the tail on the right side is longer.

$\therefore$ NOT B or D

A smaller standard deviation occurs when data is clustered more closely.

$\therefore$ NOT A where data is more widely spread.

$\Rightarrow C$

61. Statistics, STD2 S1 2007 HSC 21 MC

5, 6, 11, x , 13, 18, 25

$$\text{Range} = 2x - 6$$

$$25 - 5 = 2x - 6$$

$$2x = 26$$

$$x = 13$$

$$\therefore \text{Median} = 13$$

$$Q_1 = 6 \quad Q_3 = 18$$

$$\therefore \text{IQR} = 12$$

$\Rightarrow D$

62. Statistics, STD2 S1 2014 HSC 14 MC

Using the class centres

$$\text{Total hours} = (1 \times 5) + (4 \times 10) + (7 \times 3) + (10 \times 2)$$

$$= 5 + 40 + 21 + 20$$

$$= 86$$

$$\text{Mean hours} = \frac{86}{20} = 4.3$$

$\Rightarrow B$

◆ Mean mark 45%

COMMENT: The mean is calculated using "class centres" in grouped data.

63. Statistics, STD2 S5 2014 HSC 24 MC

Find z -scores of 2.75 and 4.15 kg

$$z(2.75) = \frac{x - \mu}{\sigma} = \frac{2.75 - 3.1}{0.35} = -1$$

$$z(4.15) = \frac{4.15 - 3.1}{0.35} = 3$$

68% between $z = -1$ and 1

$\Rightarrow$ 34% between $z = -1$ and 0

99.7% between $z = -3$ and 3

$\Rightarrow$ 49.85% between $z = 0$ and 3

$\therefore$ % with z -scores between -1 and 3

$$= 34 + 49.85$$

$$= 83.85\%$$

$\therefore$ # Babies between 2.75 kg and 4.15 kg

$$= 83.85\% \times 10\,000$$

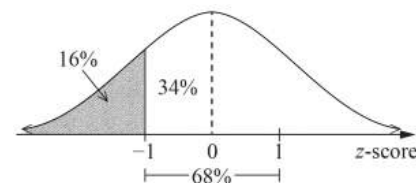
$$= 8385$$

$\Rightarrow D$

64. Statistics, STD2 S5 2015 HSC 20 MC

$$\mu = 8 \text{ cm} \quad s = 0.04 \text{ cm}$$

$$\begin{aligned} z\text{-score}(7.96) &= \frac{x - \mu}{\sigma} \\ &= \frac{7.96 - 8}{0.04} \\ &= -1 \end{aligned}$$



$\therefore$ % of pipes with a diameter less than 7.96 cm.

$$= 50\% - 34\%$$

$$= 16\%$$

$\Rightarrow A$

◆ Mean mark 50%.

65. Statistics, STD2 S1 2016 HSC 19 MC

Since range is 10 $\Rightarrow$ Last data point = 12

$$Q_1 = 3$$

$$Q_3 = \frac{8 + 9}{2} = 8.5$$

$$\text{Median} = 5$$

$\Rightarrow A$

◆ Mean mark 46%.

66. Statistics, STD2 S1 2016 HSC 21 MC

Using the centre of each class interval:

$$\text{Mean} = \frac{3 \times 3 + 8 \times 6 + 13 \times 8 + 18 \times 9}{3 + 6 + 8 + 9}$$

$$= 12.42 \dots$$

$\Rightarrow D$

◆ Mean mark 43%.

67. Statistics, STD2 S1 2018 HSC 6 MC

Data is skewed.

Since the "tail" is on the left hand side, the data is negatively skewed.

$\Rightarrow C$

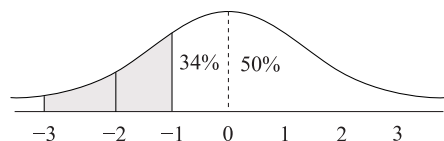
◆ Mean mark 43% (a surprisingly poor result!)

68. Statistics, STD2 S5 2018 HSC 23 MC

Given $\mu = 48$, $\sigma = 3$

$$\begin{aligned} z\text{-score (39)} &= \frac{x-\mu}{\sigma} \\ &= \frac{39-48}{3} \\ &= -3 \end{aligned}$$

$$\begin{aligned} z\text{-score (45)} &= \frac{45-48}{3} \\ &= -1 \end{aligned}$$



♦ Mean mark 47%.

$\therefore$ Scores between 39 and 45
 $\approx 16\%$

$\Rightarrow A$

Note that % of scores below 39 is very small.

69. Statistics, STD2 S1 SM-Bank 2 MC

Mean has increased (Y9 to Y12)

The Year 12 data is more tightly distributed
around the mean.

$\therefore$ Standard deviation has decreased (Y9 to Y12)

$\Rightarrow A$

70. Statistics, STD2 S1 2015 HSC 19 MC

In 1975, her life expectancy

$$\begin{aligned} &= \text{age} + \text{remaining years} \\ &= 45 + 34 \\ &= 79 \end{aligned}$$

In 1995, her life expectancy

$$\begin{aligned} &= 65 + 19.8 \\ &= 84.8 \end{aligned}$$

$$\begin{aligned} \therefore \text{Difference} &= 84.8 - 79 \\ &= 5.8 \text{ years} \end{aligned}$$

$\Rightarrow D$

♦ Mean mark 39%.

71. Statistics, 2ADV S3 SM-Bank 6 MC

$$\begin{aligned} \int_{3\pi}^a \frac{1}{4} \cos\left(\frac{x}{2}\right) dx &= \frac{\sqrt{3}+2}{4} \\ \left[\frac{1}{2} \sin\left(\frac{x}{2}\right) \right]_{3\pi}^a &= \frac{\sqrt{3}+2}{4} \\ \frac{1}{2} \left[\sin\left(\frac{a}{2}\right) - \sin\left(\frac{3\pi}{2}\right) \right] &= \frac{\sqrt{3}+2}{4} \\ \frac{1}{2} \sin\left(\frac{a}{2}\right) + \frac{1}{2} &= \frac{\sqrt{3}+2}{4} \\ \sin\left(\frac{a}{2}\right) &= \frac{\sqrt{3}}{2} \\ \frac{a}{2} &= \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \dots \\ \therefore a &= \frac{14\pi}{3} \quad (3\pi \leq a \leq 5\pi) \end{aligned}$$

$\Rightarrow B$

72. Statistics, 2ADV S3 SM-Bank 7 MC

$$\begin{aligned} \int_0^m \frac{1}{12} (8x-x^3) dx &= 0.5 \\ \left[\frac{1}{12} \left(4x^2 - \frac{x^4}{4} \right) \right]_0^m &= 0.5 \\ 4m^2 - \frac{m^4}{4} &= 6 \\ 16m^2 - m^4 &= 24 \\ m^4 - 16m^2 + 24 &= 0 \end{aligned}$$

$\Rightarrow D$

73. Statistics, STD2 S1 2020 HSC 7 MC

Positive skew occurs when the tail on the
histogram is longer on the right-hand
(positive) side.

$\Rightarrow A$

♦♦ Mean mark 32%.

74. Statistics, STD2 S1 2009 HSC 21 MC

If $\bar{x}$ of 10 scores = 14
 $\Rightarrow$ Sum of 10 scores = $10 \times 14 = 140$
With 2 additional scores, $\bar{x} = 16$
 $\Rightarrow$ Sum of 12 scores = $12 \times 16 = 192$
 $\therefore$ Value of 2 extra scores = $192 - 140$
 $= 52$
 $\therefore$ Mean of 2 extra scores = $\frac{52}{2} = 26$
 $\Rightarrow D$

◆◆◆ Mean mark 28%.

75. Statistics, STD2 S1 2010 HSC 21 MC

Since the area in the profit graph is constant and then gradually decreases after a point
 $\Rightarrow B$

◆◆◆ Mean mark 24%
COMMENT: Area graph questions have proven very challenging over recent years. Review this area.

76. Statistics, STD2 S5 2013 HSC 20 MC

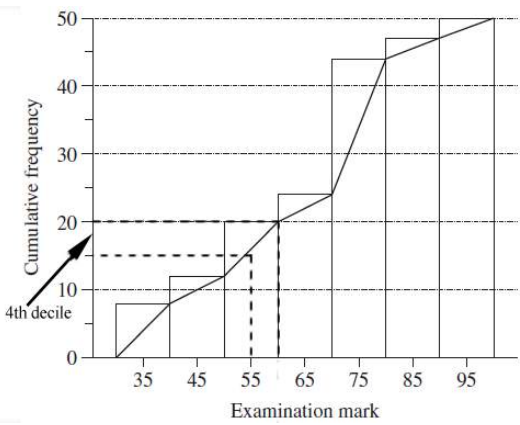
Since 68% between $z = 1$ and -1
 $\Rightarrow$ 34% between $z = 0$ and 1
Likewise, 95% between $z = 2$ and -2
 $\Rightarrow$ 47.5% between $z = 0$ and 2
 $\therefore$ % between $z = 1$ and 2
 $= 47.5 - 34$
 $= 13.5\%$
 $\therefore$ Students between $z = 1$ and $z = 2$
 $= 13.5\% \times 60\,000$
 $= 8100$
 $\Rightarrow A$

◆◆ Mean mark 24%

77. Statistics, STD2 S1 2005 HSC 22 MC

Option A: 50% of Under 21 group earned over \$325 and 75% of Over 21 group did. NOT TRUE.
Option B: 75% of Under 21 group earned below \$350 is TRUE.
Options C and D: can both be proven to be untrue using their median and quartile values.
 $\Rightarrow B$

78. Statistics, STD2 S1 2007 HSC 22 MC



4th decile occurs when cumulative frequency is between 15 and 20.
 $\therefore$ Examination mark must be between 55 and 60.
 $\Rightarrow B$

79. Statistics, STD2 S1 2016 HSC 7 MC

Categorical and nominal data is qualitative and not ordered.
 $\Rightarrow A$

◆◆ Mean mark 26%.

80. Statistics, STD2 S1 2019 HSC 10 MC

Train or bus delay (%)
 $= 92 - 86$
 $= 6\%$
 $\Rightarrow A$

◆◆◆ Mean mark 18%.

81. Statistics, STD2 S5 2019 HSC 15 MC

68% of marks lie between 64 and 76 (mean $\pm 1\sigma$).
50% of marks lie between Q_1 and Q_3 .
 $\Rightarrow A$

◆◆◆ Mean mark 17%.